

Addendum: Hybrid Knowledge Strategy

Integrating Karpathy LLM Wiki Patterns with Vectorize (RAG)

1. Knowledge Tiering Architecture

This design distinguishes between **Synthesis (The Brain)** and **Retrieval (The Library)** to solve for multi-hop reasoning and long-term memory consistency.

Component	Technology	Role	Logic Pattern
LLM Wiki	R2 (Markdown)	The "Brain": Synthesized identity and methodology.	Compilation: State is integrated and interlinked.
Vectorize Index	Cloudflare Vectorize	The "Library": Raw scientific data and archives.	Retrieval: Chunks are searched on-demand.

2. The Karpathy Wiki Pattern (R2)

Unlike RAG, which retrieves fragmented chunks, the LLM Wiki builds a compounding knowledge base where information is pre-digested by the model.

Key Markdown Components:

- `Athlete_Profile.md`: Interlinked summary of injuries, personality traits, and psychological limiters.
- `Methodology.md`: The "Coaching Bible" containing specific concurrent training protocols and tone guidelines.
- `Progress_Compendum.md`: A model-generated synthesis of D1 workout data over monthly increments.

Architectural Benefit: Enables "Multi-hop" reasoning. The agent can link a chronic injury page directly to a planned session logic without relying on semantic similarity matches.

3. Implementation: The Compilation Skill

```
// Skill: compile_session_to_wiki.ts export async function run(sessionData, currentWiki) {  
  // Instead of just storing logs, the agent REVISES the existing Wiki const updatedPage =  
  await agent.prompt("Integrate today's session into the Athlete Profile. Resolve any  
  contradictions.", { context: { sessionData, currentWikiPage:
```

```
currentWiki.get('Athlete_Profile.md') } })); await env.BUCKET.put('users/athlete_id/  
Athlete_Profile.md', updatedPage); return "Wiki compiled and state updated."; }
```

4. Auditability and Human Oversight

The Wiki pattern provides a human-readable "State of the Agent." The Supervisor Loop audits the Markdown files rather than opaque vector embeddings, allowing for surgical correction of the Coach's knowledge base.